Andrea Galenda

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PROFILE

Mechanical Engineering graduate from Politecnico di Milano (MSc, 110/110 cum laude) and former Visiting Student Researcher at NASA Jet Propulsion Laboratory, specializing in multi-physics dynamics, robotics, control and state estimation. Experienced in modeling complex mechanical systems for space applications and highly motivated to pursue a career in field robotics within international engineering environments.

EDUCATION

Politecnico di Milano Sept. 2023 – Oct. 2026
MSc. in Mechanical Engineering (Track: Mechatronics and Robotics) Milan, Italy

- **Final grade**: 110/110 cum laude
- **Thesis**: *Modelling, Dynamics and In-Flight Leakage Detection for a Variable Altitude Venus Aerobot*, developed at NASA JPL
- **Relevant coursework**: Optimal Control and State Estimation, Advanced Dynamics of Mechanical Systems, Advanced Machine Design, Actuating Devices, Signal Processing

Polytech Paris-Saclay Sept. 2024 – Feb. 2025
Erasmus+ Program, Electronics and Robotics Engineering Paris, France

Politecnico di Milano Sept. 2020 – Sept. 2023
BSc. in Mechanical Engineering Milan, Italy

EXPERIENCE

NASA Jet Propulsion Laboratory Dec. 2025 – Aug. 2026
Visiting Student Researcher - Aerial Robotic Systems (Section 358M) Pasadena (CA), United States

- Developed a high-fidelity 6-DOF flight dynamics and thermodynamics simulator in MATLAB for a variable-altitude Venus Aerobot to predict trajectory and altitude control performance
- Integrated flight dynamics, envelope shape, a six-node thermal network, and gas transport into a single fifteen-state coupled system to model a Venus aerobot over full solar day missions
- Engineered a data-driven Reduced Order Model (ROM) surrogate using Physics-Informed Neural Networks (PINNs) to rapidly predict the dynamic shape of the Balloon-in-Balloon configuration
- Implemented an Unscented Kalman Filter (UKF) leveraging existing avionics (barometer, IMU, thermistors) to perform state estimation and isolate specific-force deficits
- Designed a sequential change detection pipeline using a Cumulative Sum (CUSUM) test and constrained regression to detect leaks from 1g/d to 100g/d and separate them from sulfuric acid deposition

ACADEMIC PROJECTS

SayCan Framework for Pick & Place

- Built a SayCan framework enabling a robotic arm to perform autonomous pick-and-place operations using high-level natural language instructions, combining LLMs with vision-language detection models (ViLD).
- Fine-tuned CLIPort for open-vocabulary object grasping and implemented a dynamic feedback system with custom geometric evaluation functions to verify physical success of tasks.

Playground Safety & Floor Design: Deep Learning ROM

- Formulated parametric static structural and 2D elastodynamics problems in FEniCS to compute displacement of an elastic pad with complex internal void structures and varying layer thicknesses.
- Trained data-driven Reduced Order Models (POD-NN and DL-ROM) to bypass expensive Finite Element solves, achieving real-time predictions with sub-2.5% space-time L2 relative error.

Operational Modal Analysis for Health Assessment of Tie Rods

- Reconstructed modal parameters and Frequency Response Functions from natural excitation (NExT) data using the FDD algorithm

- Analyzed test data to evaluate dynamic behavior, isolating modal parameter variations due to structural damage from thermal and environmental disturbances

Mechanism Design and Robotics

- Designed and 3D printed a custom support and electrically actuated end-effector to manipulate the opening mechanism of a modular helmet
- Implemented robot path planning in Python, verifying autonomous task completion via camera and torque feedback

Modal Analysis

- Performed Modal Analysis on a rail wheel using frequency domain impulse test data and a least-square error minimization fitting procedure
- Conducted FEA on a harbor crane to reconstruct modal parameters, mode shapes, and its static and dynamic response under loads

Bayesian Optimization for Robot Controller Tuning

- Implemented a two-phase Bayesian Optimization strategy combining Lower Confidence Bound (LCB) exploration and Expected Improvement Plus (EI+) exploitation.
- Automated the tuning of PID control parameters and estimated equivalent link-mass parameters for a multi-joint robotic system to optimize trajectory tracking under hardware torque limits.

SKILLS

Simulation and Physics: MATLAB/Simulink, Python, Flight physics modeling, System dynamics, ROS, Gazebo
Data Analysis and Scripting: Python, C++, Signal Processing
CAD/CAE: SolidWorks, CATIA, Autodesk Inventor, Onshape, Solid Edge
Languages:English (C1), French (B2 - Reading-Listening-Speaking), Italian (Native)

EXTRACURRICULARS & VOLUNTEERING

Competitive Rugby Player (Serie A Italy, Fédérale 3 France) 2016 – 2025

- Balanced a demanding international academic career with rigorous training schedule, developing strong cross-cultural teamwork and resilience in high-pressure environments

Volunteer Support Worker June 2019 – July 2019

- Assisted and supported individuals with disabilities in a daycare center setting
- Fostered inclusive environments, developing strong interpersonal communication, patience, and empathy outside of an academic setting